CivicShield AI

AI-Powered Smart Civic Complaint & Government Response System

Hackathon Project – Detailed Concept & Prototype Plan

1. Problem Statement

In cities and villages, citizens regularly face problems such as potholes and damaged roads, non-functional streetlights, water leakage, garbage overflow, public toilet issues, fallen trees, open drainage/manholes, illegal dumping or construction, and damaged traffic signals.

Currently, complaints may be registered through different channels, but the major problems are: manual verification, wrong department assignment, delays, and lack of proper follow-up. Citizens often do not know which department is responsible, whether the complaint is being processed, how long it will take, or why it is delayed.

2. Solution

CivicShield AI creates a single intelligent platform connecting citizens and government departments.

Citizen → Upload photo/video + location → AI analyses complaint → Identifies problem + severity → Automatically assigns department → Government officer receives task → Officer resolves issue → Uploads after-repair evidence → AI verifies resolution → Citizen confirms → Complaint closed.

3. AI Module 1 – Image Classification

Citizen uploads a photo. AI identifies the civic issue and maps it to the appropriate department.

Examples: damaged road → Pothole; garbage pile → Waste Management; broken streetlight → Electrical Department; leaking pipeline → Water Department.

4. AI Module 2 – Severity Prediction

The system assigns a priority score based on issue type, location, previous complaints, nearby schools/hospitals, traffic density, population, and risk level.

Example: Open manhole + school nearby = Critical priority.

5. AI-Based Location Intelligence

GPS captures the complaint location. The system identifies the Ward → Zone → Municipality → Responsible Department and automatically routes the complaint.

6. Government Dashboard

The dashboard provides a real-time overview of total complaints, pending complaints, in-progress complaints, resolved complaints, critical/high-priority complaints, average resolution time, and map-based complaint visualization.

Map indicators: Critical, High, Medium, Resolved.

7. SLA Monitoring

Different complaint categories receive different resolution deadlines. Example targets: Open manhole – 2 hours; major water leakage – 4 hours; streetlight – 24 hours; garbage overflow – 12 hours; small pothole – 72 hours.

The system monitors SLA consumption and sends warnings before deadlines. If the SLA expires, the complaint is automatically escalated.

8. Escalation System

Example: A water leakage complaint is assigned to a Junior Engineer. If it is not resolved within the defined period, it is escalated to the Assistant Engineer and subsequently to a higher authority such as the Municipal Commissioner.

9. AI Resolution Verification

This is a key differentiating feature. The officer uploads an after-repair image. AI compares the before and after images to determine whether the reported issue appears to have been resolved.

If the repair is satisfactory, the system can mark it as verified and request citizen confirmation. If the after-image does not show a proper fix, the complaint remains open or is flagged for review.

10. Citizen Application

Home: Report Issue, Track Complaint, My Complaints.

Report Issue: Photo + GPS location + optional voice description. AI generates issue type, severity, responsible department, location, and expected resolution time.

11. Voice Complaint

Citizens can speak their complaint instead of typing. Speech-to-text converts the voice into text and AI identifies the issue, priority, and department. Example: Tamil speech about a streetlight not working can become a Streetlight Failure complaint.

12. Multilingual Support

Recommended languages: Tamil, English, and Hindi. Example: a Tamil complaint such as "■■■■■■■■■■■■ ■■■■ ■■■■■■■■■■■■" can be understood and classified as a Streetlight Complaint.

13. Government Analytics

Ward-wise complaint counts and department performance can be displayed. Example departments: Road, Water, Electrical, and Waste Management.

This helps authorities identify high-problem areas and departments with slower response times.

14. Predictive Analytics

The platform should move beyond solving existing complaints and predict future problems using historical data.

Example: If a ward has many water-leak complaints over several months, the AI can predict a high probability of pipeline failure and recommend preventive maintenance.

15. Overall Architecture

CITIZEN → Mobile/Web Application → Photo + GPS + Voice → AI Engine → Image AI + NLP AI + Severity Prediction → Complaint Engine → Department Assignment → Government Dashboard → Officer Assignment → Field Resolution → AI Resolution Verification → Resolved / Escalate

16. Recommended Technology Stack

Frontend: Flutter

Backend: Spring Boot

Database: PostgreSQL

AI: Python, OpenCV, YOLO, TensorFlow/PyTorch, NLP models

AI API: FastAPI

Integration: Spring Boot → FastAPI → AI prediction

17. Suggested Database Tables

users, complaints, departments, officers, wards, locations, complaint_images, status_history, escalations, notifications

18. Optional IoT Expansion

Water: leakage detection sensors. Garbage: ultrasonic bin-fill sensors. Streetlight: current sensors. Drainage: water-level sensors.

These devices can feed data into CivicShield AI so that IoT → CivicShield AI → Government Dashboard becomes an integrated smart-city civic management platform.

19. Government Benefits

Reduced response time; reduced duplicate complaints; better department allocation; improved transparency; faster emergency response; preventive maintenance; data-driven government planning.

20. What Makes CivicShield Different?

Existing systems generally focus on complaint registration and tracking. CivicShield should emphasize AI-based complaint classification, predictive severity, computer vision, automatic routing, SLA monitoring, escalation, and AI-based resolution verification.

Do not claim that CivicShield is the first grievance platform. Position it as an intelligent predictive and accountable layer that improves existing civic grievance workflows.

21. Recommended Hackathon MVP

Build five features strongly: (1) Citizen complaint with photo + GPS, (2) AI issue detection and severity, (3) automatic department allocation, (4) government dashboard with live complaints and map, and (5) before/after image-based resolution verification.

22. Live Demo Scenario

Upload a pothole image → system detects Pothole → Severity: Critical → Location: Ward 18 → Department: Road Maintenance → SLA: 4 hours → government dashboard receives complaint → officer changes Pending → In Progress → Resolved → officer uploads after-repair image → AI verifies resolution → citizen receives a resolution notification.

This 2–3 minute end-to-end demonstration is the strongest way to present the concept during judging.

Key Positioning for the Hackathon

CivicShield AI is not merely a complaint-registration application. It is an AI-powered civic response and accountability platform that helps governments detect, prioritize, route, monitor, verify, and predict civic issues.